



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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CSE - Academic Year 2026-27

Java Programming Lab | Batch 2025-29

ASSIGNMENT : 19

Name of Student	Anurag thakur
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Batch	CSE A - 2
Class	Java Programming Lab
Academic Year & Semester	2025-29 : Semester 3
Date of Submission	

TITLE : Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives : To establish database connectivity and retrieve records using JDBC SELECT statements.

Theory :

Java Database Connectivity (JDBC) is a standard Java API that enables Java applications to communicate with relational databases. It allows a Java program to connect to databases such as MySQL and execute SQL queries.

The **Connection** object establishes a connection with the database, while the **Statement** object is used to execute SQL statements. The **SELECT** statement retrieves records from a database table. The retrieved records are stored in a **ResultSet** object, which allows the data to be accessed row by row.

Exercise :

1. Create a Java application to display **student records from database** using SELECT query.

```

import java.sql.*; public class java19a {      public static
void main(String[] args) {      try {

Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/jdbc_lab",
        "root",
        "anurag"
    );

    System.out.println("Database successfully connected..");

    Statement st = con.createStatement();
    ResultSet rs = st.executeQuery("select * from student");

    System.out.println("Student Records");
    System.out.println("-----");
while (rs.next()) {      System.out.println(
rs.getInt("roll_no") + "  " +
rs.getString("name") + "  " +
rs.getString("course") + "  " +
rs.getInt("marks")      );
        }      con.close();
    } catch
(Exception e) {
        System.out.println(e);
    }
}
}

```

In SQL :

```
1  create database jdbc_lab;
2  use jdbc_lab;
3
4  create table student (
5      roll_no int,
6      name varchar(50),
7      course varchar(50),
8      marks int
9  );
10
11 insert into student values
12 (101, 'Anurag', 'CSE', 95),
13 (102, 'Anwesha', 'CSE', 82),
14 (103, 'Natasha', 'IT', 91);
15
--
39 select * from student;
```

Result Grid

	roll_no	name	course	marks
▶	101	Anurag	CSE	95
	102	Anwesha	CSE	82
	103	Natasha	IT	91

OUTPUT :

Database successfully connected..

Student Records

101 Anushree CSE 95

102 Anwesha CSE 82

103 Natasha IT 91

2. Create a Java application that connects to a database and displays **product details** using a **SELECT query**. Show product ID, product name, quantity, and price.

```
import java.sql.*; public class java19b {
public static void main(String[] args) {
try {
    Connection con =
DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/jdbc_lab",
        "root",
        "anurag"
    );

    System.out.println("Database successfully connected..");

    Statement st = con.createStatement();
    ResultSet rs = st.executeQuery("select * from product");

    System.out.println("Product Details");
    System.out.println("-----");
while (rs.next()) {
    System.out.println(
rs.getInt("product_id") + " " +
rs.getString("product_name") + " " +
rs.getInt("quantity") + " " +
rs.getDouble("price")
    );
    }
    con.close();
} catch (Exception e) {
    System.out.println(e);
}
}
```

In SQL :

```
16 • create table product (  
17     product_id int,  
18     product_name varchar(50),  
19     quantity int,  
20     price double  
21 );  
22  
23 • insert into product values  
24     (1, 'Laptop', 10, 65000),  
25     (2, 'Keyboard', 25, 1200),  
26     (3, 'Mouse', 30, 700);  
27  
41 • select * from product;
```

Result Grid					Filter Rows:		Export:	Wrap Cell Content:
	product_id	product_name	quantity	price				
▶	1	Laptop	10	65000				
	2	Keyboard	25	1200				
	3	Mouse	30	700				

OUTPUT :

```
Database successfully connected..  
Product Details  
-----  
1 Laptop 10 65000.0  
2 Keyboard 25 1200.0  
3 Mouse 30 700.0  
PS C:\Users\softlink\Desktop\java code>
```

Outcome :

- Understood database connectivity using JDBC.
- Learned to establish a connection between Java and MySQL.
- Implemented SELECT queries to retrieve records from database tables.
- Learned to access retrieved records using the ResultSet object.

Conclusion :

This assignment demonstrated database connectivity in Java using JDBC. The student and product records were successfully retrieved from the database using SELECT queries and displayed using the **ResultSet** object. This helped in understanding the basic process of retrieving database records using Java.